Table 8: Few-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	few-shot CoT accuracy	few-shot DA accuracy
AGIEval LSAT AR	Soft Reasoning	GPT-4o Mini	28.7	26.1
AGIEval LSAT AR	Soft Reasoning	Gemini 1.5 Flash	28.3	20.4
MMLU	Knowledge	Llama 2 7b	49.0	42.8
MMLU	Knowledge	Mistral 7b	63.0	57.0
MMLU	Knowledge	Llama 3.1 8b	71.7	69.3
MMLU	Knowledge	Llama 3.1 70b	84.3	83.7
MMLU	Knowledge	Gemma 2 9b	74.7	72.4
MMLU	Knowledge	Phi-3 Small 8k	77.3	75.2
MMLU	Knowledge	Qwen 2 7b	69.9	68.6
MMLU	Knowledge	Qwen 2 72b	82.7	81.8
MMLU	Knowledge	GPT-4o Mini	82.3	77.8
MMLU	Knowledge	Gemini 1.5 Flash	78.1	79.0
StrategyQA	Commonsense	Llama 2 7b	57.9	30.9
StrategyQA	Commonsense	Mistral 7b	70.7	72.0
StrategyQA	Commonsense	Llama 3.1 8b	74.4	65.8
StrategyQA	Commonsense	Llama 3.1 70b	87.1	84.2
StrategyQA	Commonsense	Gemma 2 9b	77.1	73.3
StrategyQA	Commonsense	Phi-3 Small 8k	75.0	71.1
StrategyQA	Commonsense	Qwen 2 7b	71.9	58.9
StrategyQA	Commonsense	Qwen 2 72b	83.2	80.1
StrategyQA	Commonsense	GPT-4o Mini	83.0	86.2
StrategyQA	Commonsense	Gemini 1.5 Flash	77.0	80.3
ContextHub Abductive L2	Symbolic	Llama 2 7b	36.2	35.0
ContextHub Abductive L2	Symbolic	Mistral 7b	33.8	30.0
ContextHub Abductive L2	Symbolic	Llama 3.1 8b	32.7	36.1
ContextHub Abductive L2	Symbolic	Llama 3.1 70b	54.6	51.2
ContextHub Abductive L2	Symbolic	Gemma 2 9b	44.8	33.2
ContextHub Abductive L2	Symbolic	Phi-3 Small 8k	49.8	34.2
ContextHub Abductive L2	Symbolic	Qwen 2 7b	39.6	35.0
ContextHub Abductive L2	Symbolic	Qwen 2 72b	54.7	34.9
ContextHub Abductive L2	Symbolic	GPT-4o Mini	62.0	60.0
ContextHub Abductive L2	Symbolic	Gemini 1.5 Flash	48.6	47.8
ContextHub Abductive L1	Symbolic	Llama 2 7b	21.4	16.7
ContextHub Abductive L1	Symbolic	Mistral 7b	23.6	21.7
ContextHub Abductive L1	Symbolic	Llama 3.1 8b	40.0	36.1
ContextHub Abductive L1	Symbolic	Llama 3.1 70b	62.2	58.9
ContextHub Abductive L1	Symbolic	Gemma 2 9b	48.9	59.4
ContextHub Abductive L1	Symbolic	Phi-3 Small 8k	59.2	56.4
ContextHub Abductive L1	Symbolic	Qwen 2 7b	48.6	38.9
ContextHub Abductive L1	Symbolic	Qwen 2 72b	53.3	56.1
ContextHub Abductive L1	Symbolic	GPT-4o Mini	77.2	59.2
ContextHub Abductive L1	Symbolic	Gemini 1.5 Flash	79.7	68.6
MuSR Murder Mysteries	Soft Reasoning	Mistral 7b	62.0	56.4
MuSR Murder Mysteries	Soft Reasoning	Llama 3.1 8b	61.6	61.2
MuSR Murder Mysteries	Soft Reasoning	Llama 3.1 70b	73.2	68.0
MuSR Murder Mysteries	Soft Reasoning	Gemma 2 9b	81.6	62.0
MuSR Murder Mysteries	Soft Reasoning	Phi-3 Small 8k	62.0	53.6
MuSR Murder Mysteries	Soft Reasoning	Qwen 2 7b	56.0	55.6
MuSR Murder Mysteries	Soft Reasoning	Qwen 2 72b	80.4	66.0
MuSR Murder Mysteries	Soft Reasoning	GPT-4o Mini	76.0	69.6
MuSR Murder Mysteries	Soft Reasoning	Gemini 1.5 Flash	70.0	66.4
MuSR Team Allocations	Soft Reasoning	Mistral 7b	42.8	43.2
MuSR Team Allocations	Soft Reasoning	Llama 3.1 8b	59.6	51.6
MuSR Team Allocations	Soft Reasoning	Llama 3.1 70b	89.2	63.6
MuSR Team Allocations	Soft Reasoning	Gemma 2 9b	48.4	45.6
MuSR Team Allocations	Soft Reasoning	Phi-3 Small 8k	66.0	46.4
MuSR Team Allocations	Soft Reasoning	Qwen 2 7b	34.0	40.8
MuSR Team Allocations	Soft Reasoning	Qwen 2 72b	56.0	66.4
MuSR Team Allocations	Soft Reasoning	GPT-4o Mini	75.6	60.0
MuSR Team Allocations	Soft Reasoning	Gemini 1.5 Flash	90.0	54.4
MMLU Pro	Knowledge	Llama 2 7b	21.5	20.4
MMLU Pro	Knowledge	Mistral 7b	34.8	26.7
MMLU Pro	Knowledge	Llama 3.1 8b	44.7	38.0
MMLU Pro	Knowledge	Llama 3.1 70b	64.4	55.1
MMLU Pro	Knowledge	Gemma 2 9b	48.5	42.4
MMLU Pro	Knowledge	Phi-3 Small 8k	54.8	43.2
MMLU Pro	Knowledge	Qwen 2 7b	46.6	39.0
MMLU Pro	Knowledge	Qwen 2 72b	62.5	51.6
MMLU Pro	Knowledge	GPT-4o Mini	63.0	45.0
MMLU Pro	Knowledge	Gemini 1.5 Flash	59.4	50.6
MuSR Object Placements	Soft Reasoning	Mistral 7b	55.5	41.0
MuSR Object Placements	Soft Reasoning	Llama 3.1 8b	66.8	50.4
MuSR Object Placements	Soft Reasoning	Llama 3.1 70b	67.2	57.4
MuSR Object Placements	Soft Reasoning	Gemma 2 9b	68.0	58.2
MuSR Object Placements	Soft Reasoning	Phi-3 Small 8k	62.1	51.6
MuSR Object Placements	Soft Reasoning	Qwen 2 7b	46.9	43.8
MuSR Object Placements	Soft Reasoning	Qwen 2 72b	66.4	43.0

Table 8: Few-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	few-shot CoT accuracy	few-shot DA accuracy
MuSR Object Placements	Soft Reasoning	GPT-4o Mini	67.0	47.0
MuSR Object Placements	Soft Reasoning	Gemini 1.5 Flash	73.0	54.7
ContextHub Deductive L2	Symbolic	Llama 2 7b	34.7	15.0
ContextHub Deductive L2	Symbolic	Mistral 7b	63.8	51.4
ContextHub Deductive L2	Symbolic	Llama 3.1 8b	76.1	27.3
ContextHub Deductive L2	Symbolic	Llama 3.1 70b	82.6	53.6
ContextHub Deductive L2	Symbolic	Gemma 2 9b	61.9	47.6
ContextHub Deductive L2	Symbolic	Phi-3 Small 8k	61.5	54.0
ContextHub Deductive L2	Symbolic	Qwen 2 7b	55.3	36.4
ContextHub Deductive L2	Symbolic	Qwen 2 72b	80.2	54.0
ContextHub Deductive L2	Symbolic	GPT-4o Mini	59.0	41.0
ContextHub Deductive L2	Symbolic	Gemini 1.5 Flash	90.2	42.5
ContextHub Deductive L1	Symbolic	Llama 2 7b	34.7	16.0
ContextHub Deductive L1	Symbolic	Mistral 7b	46.2	59.2
ContextHub Deductive L1	Symbolic	Llama 3.1 8b	73.0	23.0
ContextHub Deductive L1	Symbolic	Llama 3.1 70b	67.5	50.0
ContextHub Deductive L1	Symbolic	Gemma 2 9b	66.0	45.7
ContextHub Deductive L1	Symbolic	Phi-3 Small 8k	74.8	51.8
ContextHub Deductive L1	Symbolic	Qwen 2 7b	58.8	37.5
ContextHub Deductive L1	Symbolic	Qwen 2 72b	70.7	42.8
ContextHub Deductive L1	Symbolic	GPT-4o Mini	59.2	44.3
ContextHub Deductive L1	Symbolic	Gemini 1.5 Flash	89.3	49.8
MATH	Mathematical	Llama 2 7b	4.7	3.9
MATH	Mathematical	Mistral 7b	13.7	7.1
MATH	Mathematical	Llama 3.1 8b	41.2	14.2
MATH	Mathematical	Llama 3.1 70b	61.9	24.2
MATH	Mathematical	Gemma 2 9b	47.5	19.8
MATH	Mathematical	Phi-3 Small 8k	42.4	18.9
MATH	Mathematical	Qwen 2 7b	55.0	15.0
MATH	Mathematical	Qwen 2 72b	65.3	26.2
MATH	Mathematical	GPT-4o Mini	71.7	24.6
MATH	Mathematical	Gemini 1.5 Flash	54.7	32.3
GSM8K	Mathematical	Llama 2 7b	29.0	7.7
GSM8K	Mathematical	Mistral 7b	56.2	12.5
GSM8K	Mathematical	Llama 3.1 8b	86.4	20.1
GSM8K	Mathematical	Llama 3.1 70b	96.1	39.1
GSM8K	Mathematical	Gemma 2 9b	89.2	24.9
GSM8K	Mathematical	Phi-3 Small 8k	90.4	24.5
GSM8K	Mathematical	Qwen 2 7b	87.6	21.4
GSM8K	Mathematical	Qwen 2 72b	93.2	40.6
GSM8K	Mathematical	GPT-4o Mini	94.2	32.8
GSM8K	Mathematical	Gemini 1.5 Flash	90.6	40.4

F.3 Answer Extractor and Average Answer Span Results

In this section, we report the number of generations from each model on each dataset that our answer parser could not extract. “-1” denotes that a model was not run on a certain dataset due to context length limitations in the few-shot setting. We see that these unparseable rates are generally low across the board. The weakest models struggle on some of the most challenging datasets, but unparseable rates are all at or below 15%.

We also report the average character index of the beginning of the answer span that the answer parser extracted. Of particular note is that the direct answer prompts all return an answer within the first 60 characters, indicating that the answers are returned almost immediately, as desired. CoT completions are much longer.

G Zoom-in: MMLU and MMLU Pro

MMLU and MMLU Pro show gains from adding CoT, but because these datasets are so broad, they defy simple characterization. We explore the performance of CoT on each category of MMLU to understand divergences in CoT performance between these domains. We list the top three categories where CoT gives the largest error reduction for Llama 3.1 8B and 70B on MMLU and MMLU Pro in Table 17. Some of these categories are explicitly mathematical in nature, as we might expect from Figure 3. We can also see that CoT is helping on categories like “business”; upon closer inspection, we found that these categories frequently involve math as well (e.g., business questions may involve computations surrounding wealth). We need to more carefully characterize MMLU at the *instance*

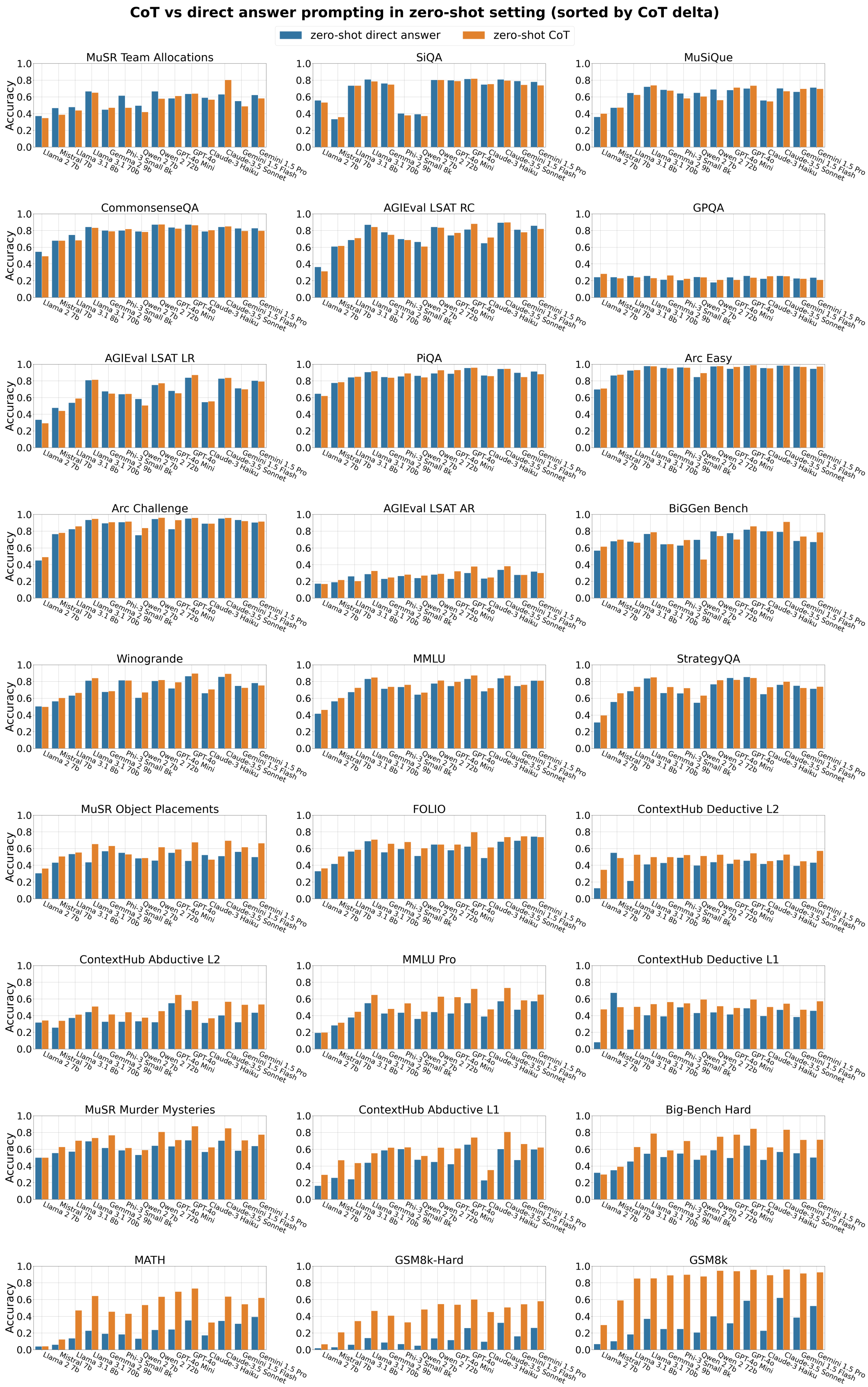

Figure 9: Performance of zero-shot direct (blue) and zero-shot CoT (orange) across datasets and models. Graphs are sorted in ascending order by median delta (CoT, direct). The datasets benefiting substantially are all symbolic or semi-symbolic in nature.